

1. List 1~2 characterization techniques to discriminate between crystalline and amorphous solid.
2. Please list the four types of bonding in solids. And give an example for each bonding and order their strengths.
3. Please list the seven basic unit cells and give the unit-cell parameters for each one (a, b, c and $\alpha\beta\gamma$)
4. The names of the two kinds of closest packing in crystals? And calculate how many particles in a primitive unit cell?
5. Please give the coordination numbers for trigonal, tetrahedral, octahedral and cubic voids.
6. Why plastic deformation occurs at stresses that are much smaller than the theoretical strength of perfect crystals?
7. Explain why the color of Fe^{2+} doped beryl crystal is blue, while the color of Fe^{3+} doped same crystal is yellow?
8. List the steps in a sol-gel process? How does the pH value affect the morphology of silica gel?

Chapter II

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1. Amorphous and crystalline solid

- Discriminating the amorphous solid from the crystalline solid.
- What characterization techniques can be used to make a determination.

2. Four types of bonding in solids

- Covalent; ionic; metallic; molecular (can order them)
- Ionic solid: comparing the melting temperatures (For example NaCl and KCl);
- Metallic solid: How to define metallic bonding? What is delocalized ions?
How to understand that metallic solids generally show high electrical/thermal conductivity and good malleability/ductility;
- Molecular solid: van der Waals forces and hydrogen bonding;
- Covalent network solid

3. Crystalline state

- How to choose the most representative unit cell for a crystal?
- Miller indices
- How to calculate the number of the motif in a unit cell? (at center, in corner, on face, on edge)
- The seven basic unit cells (names and properties)
- Crystal lattices and close packing: simple cubic; body-centered cubic; hexagonal closest-packed (hcp); face centered cubic packing (fcc).
- Interstitial crystal lattices (coordination number): trigonal; tetrahedral; octahedral; cubic.
- Understanding the crystal lattices in the following website (note fcc=ccp):
<http://www.dawgsgdk.org/crystal/en/library/wurtzite#0009>
- The difference between macroscopic and microscopic symmetry.
- The properties of crystal: hardness, cleavage, color, birefringence, piezoelectricity.
- Crystal defects: dislocation is important (Why the plastic deformation properties could be changed to a very large degree by forging without changing the chemical composition?)

4. Amorphous state

- Sol-gel processing: what is sol? What is gel?
- How to increase the colloidal stability of nanoparticles?
- The steps of sol-gel fabrication.
- How pH catalysis controls the shape of silica particles?
- Glasses: the main ingredients of glasses are sand (silica (SiO_2)_n) and soda ash (Na_2CO_3).
- Cement: the primary difference between concrete and cement.